Project:
The Slient Voice Bridge

The Team
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Meeting 4/9/2026

3-5 pm on Zoom

Introduction/Goal

Goal: Bridge the communication gap for Deaf and Hard-of-Hearing community by creating portable, high-speed ASL interpreter that runs on NVIDIA Jetson

- Capable of translation ASL into English test
- Low latency, works offline, in real-time
- No need for interpreters



Research - Dataset Problem



Dataset Type Needed:

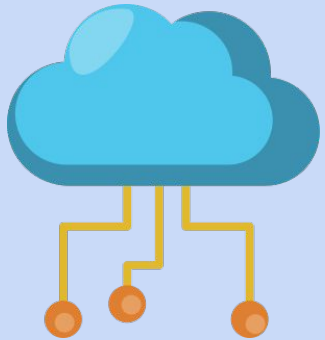
- Video-based dataset with labeled ASL signs (not static images)
- Multiple signers to capture variation in signing styles
- High frame rate (24-30 fps) for smooth motion capture

Existing Datasets:

- **WLASL** — 2,000+ signs, 21,000+ videos
- **MS-ASL** — 1,000 signs, 25,000+ videos
- **How2Sign** — continuous sentence-level signing

Additional Steps:

- Fine-tune with custom recordings
- Augment data (flips, rotations, speed variations)
    - Helps the model recognize signs even when conditions aren't perfect



Research - NVIDIA Jetson Orin Nano/FPGA

- Jetson Orin Nano dev kit contains camera-friendly interfaces including MIPI CSI connectors for vision projects.
- Camera → Jetson Orin Nano → ASL model → text/speech output
- Later in 2 chip design we have the FPGA do preprocessing like image resizing and coloring and ROI then pass to the data to Jetson to do things like neural network inference, sequence modeling and UI, sequence modeling and speech.
- Camera → FPGA → Jetson Orin Nano → output

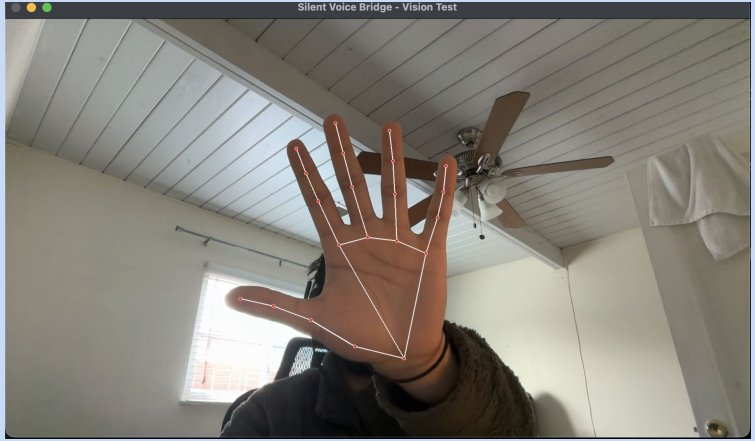
Achievement (so far…)

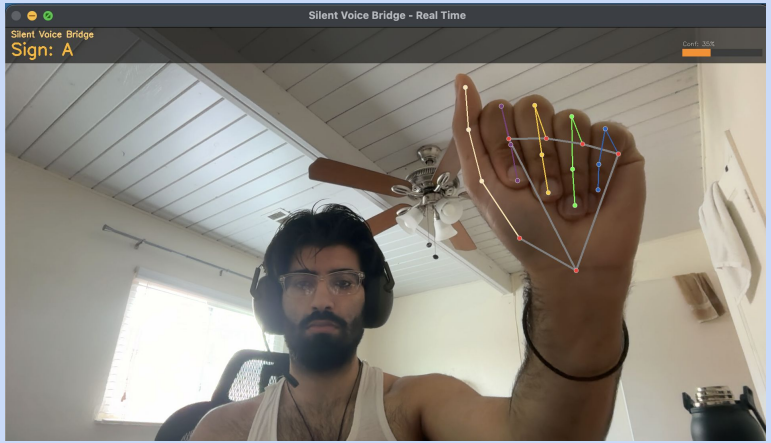
Milestone 1: Classification Foundation

- Dataset: Kaggle ASL Alphabete dataset - 87,000 images across 29 classes.

Pipeline:

- Processed 15 gb of raw images in 50MB CSV of coordinates
- Used MediaPipe to extract 21 (x,y,z) landmarks for each hand
- Implemented Normalization to ensure it works in real-time frame
- Trained on basic Random Forest Classifier





Next Steps (future goals)

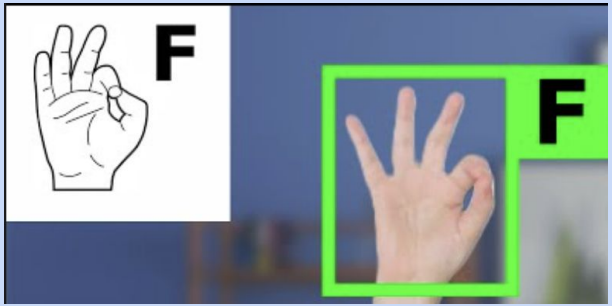
Moving from static pictures to Dynamic by using Word-Level ASL (WLASL) dataset to learn complex signs like “Help”, “Hello”, and “Thank You.”

LSTM Neural networks: Long Short-Term Memory layers to analyze sequences of frames (30-frame windows)

After our basic model can do inference and understand complex signs then…

NIVIDIA Jetson Deployment: Moving Python env into Docker Container perhaps ?





Any Feedback (from KaiKai)...

Dataset Problem

User Experience: Implement ASL to English text after visual translation accuracy is high

People who don’t know ASL, can try to learn as well…

Nvidia Jetson: Power benchmarking, FPGA…